UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5054 PHYSICS

5054/42 Paper 4 (Alternative to Practical), maximum raw mark 30

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1 (a) (i) move lens (along the ruler) / moves object and screen together (lens not moved) B1 [1]

(ii) (horizontally) align (centres) of object, lens and screen / raise object / lower lens B1 [1]
(allow raise screen)

(iii) any sensible answer for finding middle of side of block, e.g.
how non-parallax used such as viewed from above
measuring of length of block and divide by 2 B1 [1]

(b) (i) 0.14 m cao B1 [1]

(ii) 0.245(1) m allow 0.25 m B1 [1]

(c) (i) axes: labels correct way round, labelled quantity and unit B1
scales: more than ½ grid, sensible, values consistent with labels B1
2 cm ≡ 0.1 cm on both axes
points plotted accurately B1
straight line of best fit neatly drawn through all points B1 [4]

(ii) 0.97 to 1 ignore unit B1
correct use of at least half graph line ($\Delta D \geq 0.2$) shown on graph or in calculation B1 [2]

(iii) 0.24 m to 0.25 m B1 [1]

(d) (more accurate because) gradient / more readings gives **average** (of different readings) / can ignore anomalous points / straight line from many/several points B1 [1]

[Total: 13]

2 (a) (i) circuit with power supply and given wire with ammeter in series B1
variable resistor / variable power supply B1 [2]

(ii) decrease variable resistor/resistance (of variable resistor) / increase supply voltage / increase number of cells B1 [1]

(iii) reverse connections to battery/cell / change polarity of battery
(accept reverse wire in the field) B1 [1]

(iv) turn magnet other way up / S-pole on top and N-pole under wire / change polarity of magnets B1 [1]

(b) wire becomes hot / melts / fuses / burns / trips power supply / damages/fuses ammeter B1 [1]

[Total: 6]

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3 **(a)** **(i)** movement of water/purple colour/crystal clear(er)/takes longer/more visible (to class) B1 [1]

(ii) water stops moving B1 [1]

(iii) water moves slowly ora e.g. all happens too quickly B1 [1]

(b) arrow(s) up start from/above crystal B1
arrow(s) to left near bottom of water / arrow(s) down on right B1 [2]

(c) water/beaker already warm / water already coloured B1 [1]

[Total: 6]

4 **(a)** solid state detector / Geiger counter / Geiger-Muller/Geiger/GM tube B1 [1]

(b) **(i)** 53.6 / 54 / 0.447 seen / ÷120 seen / Σvalues/5 C1
0.45 cao A1 [2]

(ii) (radioactive) decay is random (in time) B1 [1]

(c) no (radiation) source / count rate low / always present (in environment) / no (additional) hazard / source is in lead box B1 [1]

[Total: 5]

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